

Preventing Cross Network Slice Disruptions in a Zero-Trust and Multi-Tenant Future 5G Networks

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Abstract

As network slicing is the chief enabler for future Beyond 5G (B5G) and 6G networks, multiple tenants interoperate cost-effectively to provide a variety of slice services on a common physical infrastructure. However, this opens the doors to cross-slice disruptions via Man-in-the-Middle (MITM) attacks, which ultimately disrupt slice services in the data plane. In this paper, the authors address possible cross-network slice disruptions in a zero-trust and multi-tenant based 5G network by proposing two complementary design techniques: (1) secure communication mechanisms and (2) Artificial Intelligence (AI)-based anomaly detection. Experiments on a 5G testbed prototype demonstrate that Attribute-Based Encryption (ABE) provides higher security benefits in confidentiality and implicit authorization, while symmetric encryption combined with integrity protection prevents cross-slice disruptions with lower communication overhead at the cost of a reduced security level.

Keywords: 5G network slicing, zero-trust architecture, multi-tenancy, cross-slice attacks, Man-in-the-Middle (MITM), Attribute-Based Encryption (ABE), AI anomaly detection, B5G, 6G security

1. Introduction

The fifth generation (5G) of mobile networks represents a fundamental architectural shift from monolithic, purpose-built infrastructure toward a dynamic, software-defined platform capable of simultaneously serving heterogeneous applications — from enhanced Mobile Broadband (eMBB) to Ultra-Reliable Low-Latency Communications (URLLC) and massive Machine-Type Communications (mMTC). The key technology underpinning this versatility is **network slicing**, which enables the creation of logically isolated, end-to-end virtual networks over a shared physical substrate.

In a multi-tenant environment, Mobile Network Operators (MNOs) lease slices to vertical industries — automotive, healthcare, industrial IoT — each with distinct Quality of Service (QoS) requirements. While this model yields significant cost efficiencies, it simultaneously introduces severe security challenges: a malicious or compromised tenant can potentially interfere with neighbor slices, violating isolation guarantees and triggering cascading service degradation.

Traditional perimeter-based security models assume a trusted internal network, an assumption that is fundamentally incompatible with the open, virtualized nature of 5G. The **Zero-Trust Architecture (ZTA)** paradigm, which mandates continuous verification of every entity regardless of network location, offers a principled basis for securing multi-tenant 5G deployments. This paper investigates cross-network slice disruption scenarios under a zero-trust lens and proposes concrete countermeasures validated on a physical 5G testbed.

2. Background and Related Work

2.1 Network Slicing Architecture

A 5G network slice is an end-to-end logical network composed of virtualized network functions (VNFs) orchestrated by the Management and Network Orchestration (MANO) framework. The 3GPP architecture distinguishes three principal slice types: eMBB for high-throughput scenarios, URLLC for latency-critical applications, and mMTC for dense IoT deployments. Each slice is identified by a Single Network Slice Selection Assistance Information (S-NSSAI) value and is expected to operate within strict isolation boundaries at both the control and data planes.

Slice isolation is enforced through a combination of mechanisms: virtual LAN segmentation at the transport layer, namespace separation in containerized environments (Kubernetes), and flow-level policies in Software-Defined Networking (SDN) controllers. However, imperfect enforcement of these boundaries — particularly at inter-slice interfaces and shared User Plane Functions (UPFs) — creates exploitable vulnerabilities.

2.2 Zero-Trust Architecture in 5G

Zero-Trust is a security philosophy articulated in NIST SP 800-207 that eliminates implicit trust and enforces strict identity verification, least-privilege access, and micro-segmentation. Applied to 5G, ZTA demands that every inter-slice and intra-slice communication be authenticated and authorized, irrespective of whether the communicating entities reside on the same physical host. Key ZTA principles include: (i) never trust, always verify; (ii) assume breach; (iii) verify explicitly using all available data points including identity, location, device health, and behavioral signals.

2.3 Threat Landscape

Prior work has identified several attack categories targeting network slicing environments. Man-in-the-Middle (MITM) attacks, wherein an adversary intercepts and potentially modifies traffic between two legitimate slice endpoints, are particularly damaging because they operate at the data-plane level, often evading control-plane security checks. Related attacks include slice hopping — where a tenant covertly accesses neighbor slice resources via shared memory or CPU contention — and resource exhaustion attacks that monopolize physical infrastructure to degrade other tenants' slices (effectively a cross-slice Denial of Service).

A critical vulnerability disclosed by AdaptiveMobile Security demonstrated that flaws in 5G core slicing logic could permit cross-slice data exfiltration and denial of service in multi-tenant deployments. Studies such as 5GLatte and 5GReplay further showed that weak input validation and poor isolation enforcement can enable lateral movement across both control and data planes, underscoring the urgency of this research area.

3. System Model and Threat Scenario

3.1 Multi-Tenant 5G Slice Model

The authors consider a 5G system comprising a shared physical infrastructure managed by an MNO. Multiple tenants (Tenant A, Tenant B, ..., Tenant N) each operate one or more network slices instantiated atop this infrastructure. Each slice contains its own VNFs — including AMF, SMF, and UPF instances — that are logically isolated but may share physical compute and transport resources. The inter-slice communication fabric is realized via an SDN-enabled transport network with programmable forwarding rules.

3.2 Adversary Model

The adversary is modeled as an internal threat: a malicious tenant (or a compromised VNF within a legitimate tenant's slice) with the capability to inject or intercept packets at the data-plane level. The attacker's goal is to execute a Man-in-the-Middle (MITM) attack against traffic belonging to a victim slice, thereby achieving one or more of: (1) confidentiality violation — reading victim slice payload; (2) integrity violation — modifying traffic to alter application behavior; (3) availability disruption — dropping or delaying victim traffic to violate SLA guarantees.

Attack Type	Target	Impact	Data Plane Effect
MITM Eavesdropping	UPF inter-slice link	Confidentiality breach	Traffic interception
MITM Tampering	Application data flow	Integrity violation	Payload modification
Traffic Injection	Victim slice GTP tunnel	Availability / integrity	Spurious packet flood
Cross-Slice DoS	Shared UPF resources	Availability degradation	Resource exhaustion
Slice Hopping	Neighbor slice namespace	Confidentiality breach	Side-channel leakage

Table 1: Cross-Slice Attack Taxonomy Considered in This Work

4. Proposed Prevention Techniques

The paper proposes two complementary defense strategies, deployed within a zero-trust framework: (A) secure communication mechanisms that cryptographically enforce slice isolation, and (B) AI-based anomaly detection that identifies cross-slice attack patterns in real time. These techniques are not mutually exclusive; rather, they are designed to complement each other in a layered defense architecture.

4.1 Secure Communication Mechanisms

4.1.1 Attribute-Based Encryption (ABE)

Attribute-Based Encryption (ABE) is a public-key cryptographic paradigm in which ciphertext and/or decryption keys are associated with descriptive attributes (e.g., slice_id=URLLC, tenant=TenantA). Decryption is possible only if the key's attributes satisfy the access policy embedded in the ciphertext, providing cryptographically enforced access control without requiring a centralized key distribution authority.

In the proposed architecture, each slice is assigned a unique set of attributes derived from its S-NSSAI and tenant identity. Data-plane packets are encrypted under an ABE scheme such that only VNFs belonging to the authorized slice can decrypt them. This provides two critical security properties: **confidentiality** (a rogue VNF in a different slice cannot decrypt intercepted packets) and **implicit authorization** (successful

decryption implicitly proves the receiver holds the correct slice identity).

The trade-off of ABE is computational overhead: pairing-based cryptography used in most practical ABE schemes incurs higher latency compared to symmetric alternatives. The experimental evaluation quantifies this overhead relative to the security benefits.

4.1.2 Symmetric Encryption with Integrity Protection

As a lighter-weight alternative, the authors evaluate symmetric encryption (AES in GCM mode) combined with message authentication codes (MAC). Per-slice keys are provisioned via the MANO framework at slice instantiation and rotated periodically. The GCM mode simultaneously provides authenticated encryption, ensuring both confidentiality and integrity in a single cryptographic operation with negligible overhead compared to AES in CTR mode alone.

While this approach substantially reduces cross-slice eavesdropping and tampering, it provides weaker authorization guarantees than ABE: a compromised key holder within the legitimate slice could still misuse credentials. The per-slice key isolation partially mitigates this risk by limiting the blast radius of a key compromise to a single slice.

4.2 AI-Based Anomaly Detection

In parallel with cryptographic controls, the authors deploy an AI-based anomaly detection module at the SDN controller plane. This module continuously monitors per-slice traffic telemetry — including packet inter-arrival times, byte counts, flow duration, and GTP tunnel metadata — and learns baseline behavioral profiles for each slice.

The anomaly detection engine employs a combination of statistical models and machine learning classifiers trained on both benign and adversarial traffic traces. Features are extracted from the telemetry stream and fed to a lightweight classifier (e.g., Isolation Forest or LSTM-based sequence model) that outputs an anomaly score for each active flow. Flows exceeding a tunable threshold trigger an alert and optionally invoke automated mitigation (flow quarantine or rerouting) via the SDN southbound API.

The AI module is designed to operate at line rate with minimal CPU overhead by leveraging streaming telemetry via gRPC/gNMI interfaces from the SDN data plane, avoiding the latency penalty of packet-level deep inspection.

5. Experimental Evaluation

5.1 Testbed Setup

Experiments were conducted on a 5G testbed prototype comprising: (i) a 5G core network (Open5GS) deployed in containerized VNFs on Kubernetes; (ii) an Open-RAN-compatible RAN emulator (UERANSIM) simulating multiple UEs across different slices; (iii) an SDN controller (ONOS) managing the inter-slice transport fabric; and (iv) a traffic generator producing realistic eMBB and URLLC slice workloads. The MITM adversary was emulated by injecting a malicious VNF at a shared network segment between two tenant slices.

Metric	No Protection	Sym. Enc. + MAC	ABE
Cross-Slice Confidentiality	Violated	Protected	Protected

Implicit Authorization	None	None	Enforced
MITM Tamper Detection	None	Detected (MAC fail)	Detected (decrypt fail)
Encryption Overhead (ms/pkt)	—	~0.08	~2.4
Key Management Complexity	Low	Moderate	High
Security Level	None	Medium	High

Table 2: Comparative Evaluation of Cross-Slice Protection Mechanisms

5.2 Key Results

Secure Communication: ABE provided complete confidentiality protection against the MITM adversary: all intercepted packets were unreadable without the correct slice attributes, and implicit authorization prevented replay across slices. However, the pairing-based cryptographic operations introduced a per-packet overhead of approximately 2.4 ms, making ABE primarily suitable for control-plane or infrequent data-plane operations where latency tolerance is higher.

Symmetric Encryption with MAC: AES-GCM achieved near-zero overhead (~0.08 ms/packet), making it viable for high-throughput eMBB slices. Integrity protection successfully detected all tampering attempts (MAC verification failures were triggered within one RTT of the injection event). The main limitation is the absence of implicit authorization: a key-sharing arrangement between tenants could theoretically be exploited, though per-slice key isolation significantly limits this vector.

AI Anomaly Detection: The AI module demonstrated high sensitivity to MITM-induced traffic anomalies, detecting injected flows within a detection window of fewer than five seconds in controlled experiments. False positive rates were low during steady-state operation but increased transiently during legitimate traffic bursts, pointing to the need for adaptive threshold tuning. The AI engine consumed less than 3% of controller CPU resources during peak load, confirming operational viability.

6. Discussion

The experimental results confirm that no single countermeasure is optimal across all dimensions of security and performance. ABE excels in environments where authorization granularity and confidentiality are paramount (e.g., healthcare or government slices), while symmetric encryption with MAC is the preferred choice for high-throughput, latency-sensitive applications. Combining both with AI-based anomaly detection yields a defense-in-depth posture aligned with zero-trust principles.

An important architectural insight from this work is that zero-trust enforcement in 5G must extend into the data plane, not remain confined to the control plane. Current 3GPP standards do not mandate per-slice data-plane encryption, leaving this as an operator responsibility. The proposed techniques can be integrated into existing MANO workflows without modifying standardized 5G interfaces, offering a practical deployment path.

Scalability is a recognized challenge: ABE key management complexity grows with the number of slices and tenants, necessitating efficient key distribution mechanisms (e.g., hierarchical ABE schemes or delegated authority models). Similarly, the AI detection model must be continually retrained as slice traffic patterns evolve — an area where 3GPP's AI/ML framework for network management (Release 18 onwards) provides

an emerging standardization anchor.

7. Conclusion

This paper addressed the critical and underexplored problem of cross-network slice disruptions in zero-trust, multi-tenant 5G networks. The authors formalized an adversary model targeting data-plane MITM attacks and proposed two prevention techniques — Attribute-Based Encryption and symmetric encryption with integrity protection — evaluated on a physical 5G testbed alongside an AI-based anomaly detection module.

The principal conclusions are: (1) ABE delivers superior security guarantees (confidentiality and implicit authorization) at the cost of higher cryptographic overhead; (2) symmetric AES-GCM encryption provides a practical, low-overhead alternative that prevents cross-slice disruptions with adequate but weaker security; (3) AI anomaly detection complements cryptographic defenses by detecting evasive attacks and behavioral deviations in near real time. Together, these techniques form a comprehensive zero-trust defense stack for the B5G and 6G era.

Future work will investigate scalable ABE key distribution for large-scale sliced networks, federated learning approaches for distributed AI anomaly detection across multi-domain 5G deployments, and formal security proofs for the proposed composition of defenses.

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